

El protocolo HTTP

1 Análisis de una URL

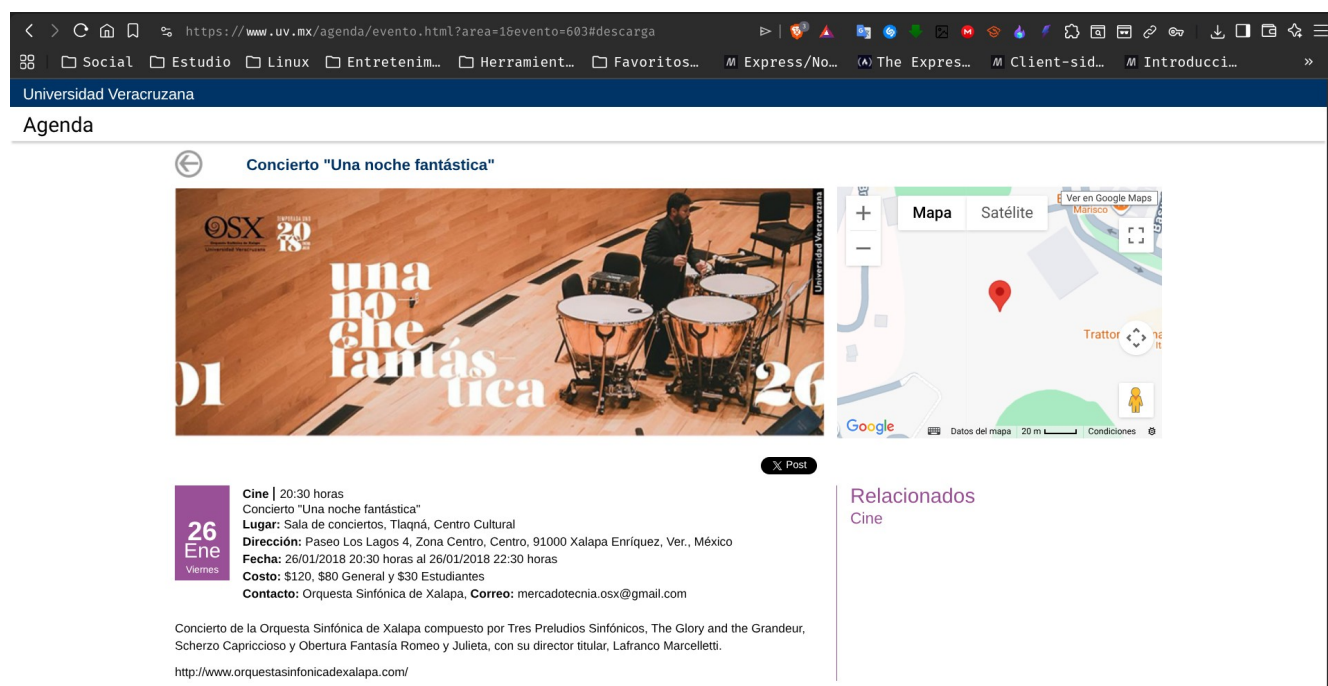
- Esquema: https
- Dominio: uv.mx
- Ruta: <https://www.uv.mx/agenda/evento.html>
- Parámetros: area=1 evento=603
- Fragmento: #descarga

- ¿Qué parte se envía al servidor?

Se envía una petición para acceder a un recurso en específico.

- ¿Qué parte solo interpreta el navegador?

El navegador interpreta únicamente un fragmento de todo el recurso, en este caso, la información del concierto “Una noche fantástica”.



Concierto "Una noche fantástica"

OSX 20 años

una noche fantástica

26 Ene Viernes

Cine | 20:30 horas
Concierto "Una noche fantástica"
Lugar: Sala de conciertos, Tlaqná, Centro Cultural
Dirección: Paseo Los Lagos 4, Zona Centro, Centro, 91000 Xalapa Enríquez, Ver., México
Fecha: 26/01/2018 20:30 horas al 26/01/2018 22:30 horas
Costo: \$120, \$80 General y \$30 Estudiantes
Contacto: Orquesta Sinfónica de Xalapa, Correo: mercadotecnia.osx@gmail.com

Concierto de la Orquesta Sinfónica de Xalapa compuesto por Tres Preludios Sinfónicos, The Glory and the Grandeur, Scherzo Capriccioso y Obertura Fantasía Romeo y Julieta, con su director titular, Lafranco Marcelletti.
<http://www.orquestasinfonicadexalapa.com/>

Relacionados
Cine

2 Observación del paradigma solicitud-respuesta

- Método HTTP: GET
- URL solicitada: <https://www.uv.mx/agenda/evento.html?area=1&evento=603>
- Código de estado: 200 la primera vez que se accede y 304 al recargar porque accede a una versión guardada en el cache.
- Dirección IP del servidor: 20.88.208.126
- Puerto utilizado: 443

- ¿Cuántas solicitudes fueron necesarias para cargar la pagina completa?

- ¿Qué tipo de recursos adicionales se descargaron?

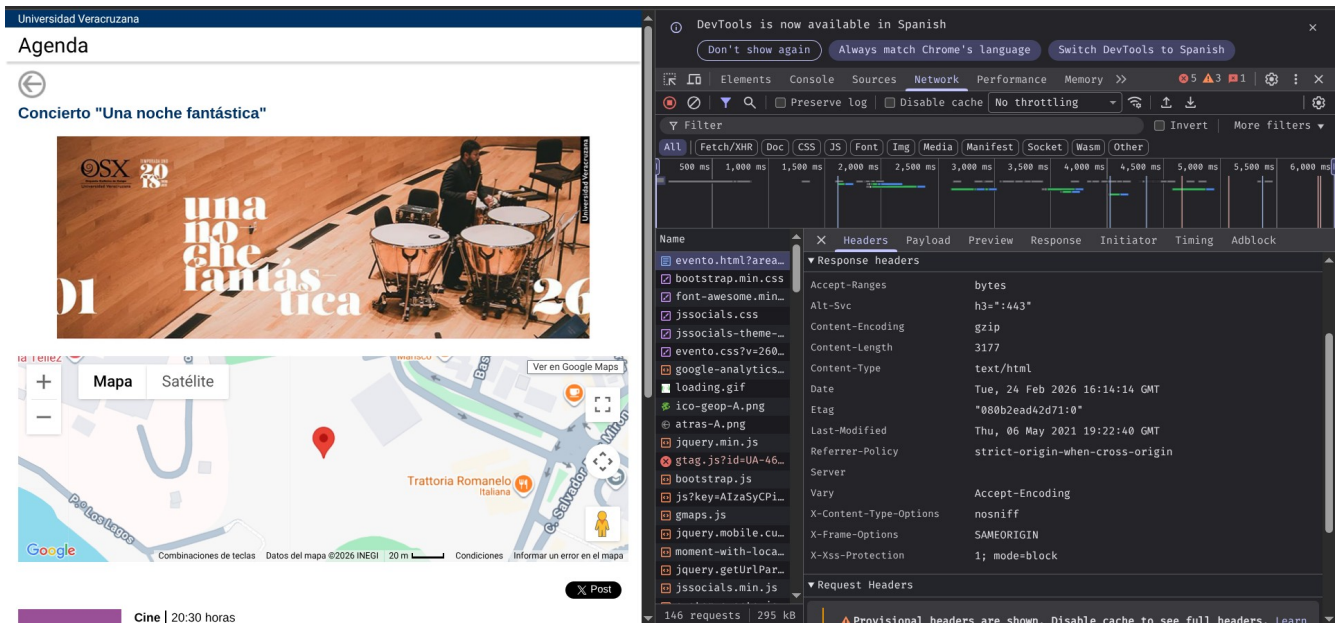
The image shows a web browser displaying the 'Agenda' page of Universidad Veracruzana. The page features a concert titled 'Una noche fantástica' with a photo of a musician. Below the photo is a map showing the location of the event. The Chrome DevTools Network tab is open, showing a list of requests. The selected request is 'StaticMapService...', which is a GET request to 'https://www.uv.mx/agenda/evento.html?area=1&evento=603'. The status is '200 OK (from disk cache)'. The response headers show 'Accept-Ranges: bytes', 'Alt-Svc: h3=":443"', 'Content-Encoding: gzip', 'Content-Length: 3177', 'Content-Type: text/html', 'Date: Tue, 24 Feb 2026 16:04:06 GMT', 'Etag: "080b2ead42d71:0"', 'Last-Modified: Thu, 06 May 2021 19:22:40 GMT', 'Referrer-Policy: strict-origin-when-cross-origin', and 'Server: Vary'. The bottom of the DevTools window shows '145 requests | 297 kB'.

3 Análisis de cabeceras HTTP

- Dentro de la misma petición, observar:

- Request headers
 - Host: `www.uv.mx`
 - User-Agent: `Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36`
 - Accept: `text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8`
 - Cookie: None
- Response headers
 - Server: None
 - Content-Type: `text/html`
 - Content-Length: `3177`
 - Date: `Tue, 24 Feb 2026 16:04:06 GMT`

- ¿Qué información envía el cliente?
- ¿Qué información expone el servidor?
- ¿Qué cabeceras podrían representar un riesgo si están mal configuradas?



4 HTTP vs HTTPS

1. En la terminal ejecuta:

```
curl -v http://neverssl.com
```

```
> curl -v http://neverssl.com
* Host neverssl.com:80 was resolved.
* IPv6: 2600:1f13:37c:1400:ba21:7165:5fc7:736e
* IPv4: 34.223.124.45
* Trying [2600:1f13:37c:1400:ba21:7165:5fc7:736e]:80 ...
* Trying 34.223.124.45:80 ...
* Connected to neverssl.com (2600:1f13:37c:1400:ba21:7165:5fc7:736e) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: neverssl.com
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 24 Feb 2026 16:29:39 GMT
< Server: Apache/2.4.62 ()
< Upgrade: h2,h2c
< Connection: Upgrade
< Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT
< ETag: "f79-5e28b29d38e93"
< Accept-Ranges: bytes
< Content-Length: 3961
< Vary: Accept-Encoding
< Content-Type: text/html; charset=UTF-8
<
<html>
  <head>
    <title>NeverSSL - Connecting ... </title>
    <style>
      body {
        font-family: Montserrat, helvetica, arial, sans-serif;
        font-size: 16px;
        color: #444444;
        margin: 0;
      }
      h2 {
        font-weight: 700;
        font-size: 1.6em;
```

2. Observar:

Cabeceras

Código de estado

Redirecciones

3. Ejecutar:

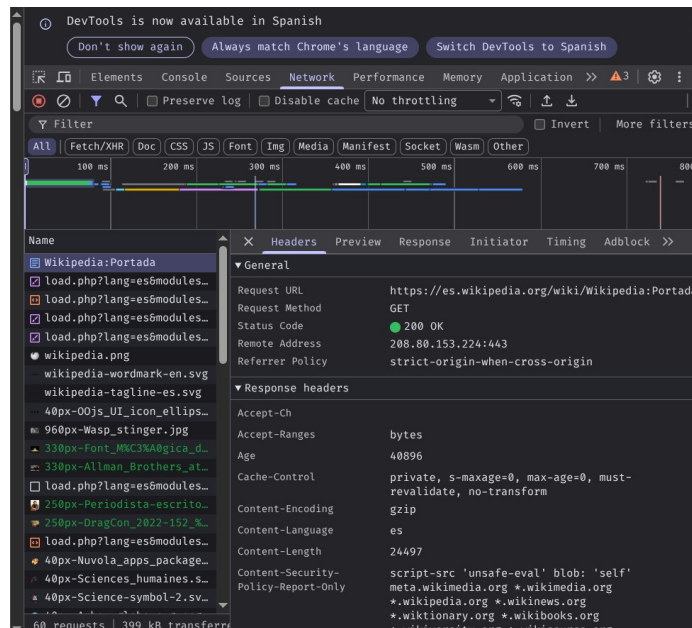
`curl -v https://neverssl.com`

```
curl: (30) Recv failure: Conexión reiniciada por la máquina remota
> curl -v https://neverssl.com
* Host neverssl.com:443 was resolved.
* IPv6: 2600:1f13:37c:1400:ba21:7165:5fc7:736e
* IPv4: 34.223.124.45
*   Trying [2600:1f13:37c:1400:ba21:7165:5fc7:736e]:443...
* Immediate connect fail for 2600:1f13:37c:1400:ba21:7165:5fc7:736e: La red es inaccesible
*   Trying 34.223.124.45:443...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CAPath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_128_GCM_SHA256 / x25519 / RSASSA-PSS
* ALPN: server did not agree on a protocol. Uses default.
* Server certificate:
*  subject: CN=neverssl.com
*  start date: Jul  9 00:00:00 2025 GMT
*  expire date: Aug  7 23:59:59 2026 GMT
*  subjectAltName: host "neverssl.com" matched cert's "neverssl.com"
*  issuer: C=US; O=Amazon; CN=Amazon RSA 2048 M04
*  SSL certificate verify ok.
*   Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 1: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 2: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Connected to neverssl.com (34.223.124.45) port 443
* using HTTP/1.x
> GET / HTTP/1.1
> Host: neverssl.com
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
* TLSv1.3 (IN), TLS handshake, NewSession Ticket (4):
* TLSv1.3 (IN), TLS alert, close notify (256):
* Empty reply from server
* shutting down connection #0
curl: (52) Empty reply from server
```

- ¿Qué diferencia se observa en la conexión? La primera resulta en una conexión exitosa mientras la segunda no devuelve nada.
- ¿Qué puerto se utiliza en cada caso? Para la primera liga (http) se conecto mediante el puerto 80 mientras la segunda liga (https) uso el puerto 443.
- ¿Qué aparece relacionado con TLS? Se muestra el proceso de handshake y sus fases; desde el “Cliente hello” hasta el verificación del cifrado.

5 Demostración de que HTTP es sin estado

1. Acceder a un sitio cualquiera

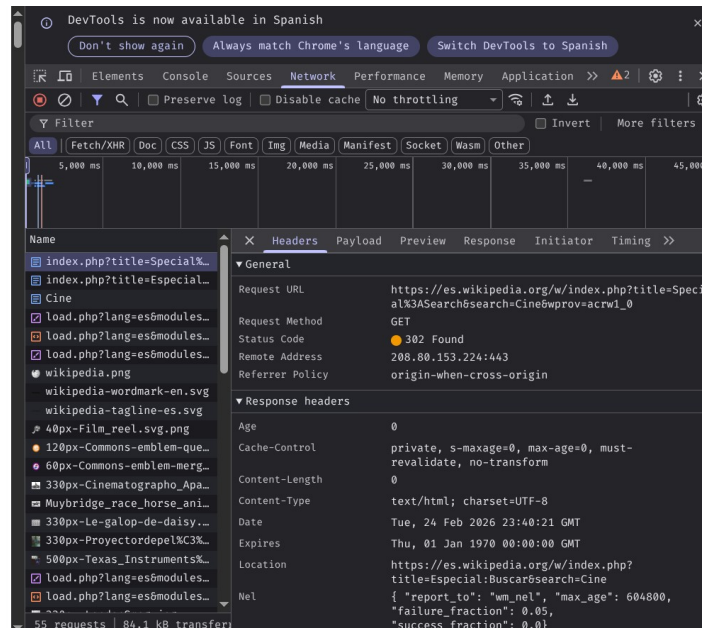


2. Realizar una solicitud

Se busca algo dentro de Wikipedia.



The screenshot shows the Wikipedia page for 'Cine'. At the top, there's a navigation bar with the Wikipedia logo, search, and links for donations, account creation, and login. Below the navigation bar, the title 'Cine' is displayed with a language selector for 4 languages. The page content includes two informational boxes: one stating that the article needs more references for verifiability, and another suggesting it be merged with the 'Cinematografía' article. Below these boxes, the text explains that 'Cine' is the abbreviation for 'cinematógrafo' or 'cinematografía', which is the technique and art of creating and projecting motion pictures. It mentions that the word was created at the end of the 19th century from two Greek words: 'kinē' (movement) and 'graphō' (drawing). To the right of the text is an image of a Lumière camera with the caption 'Cámara cinematográfica de los hermanos Lumière.'

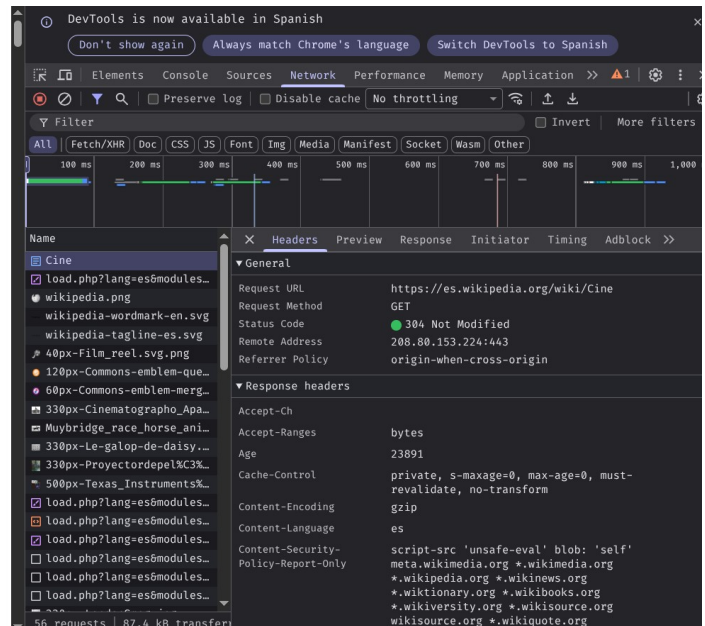


The screenshot shows the Chrome DevTools Network tab. It displays a list of network requests, with the first one selected: 'index.php?title=Special:Buscador&search=Cine&prov=acrwl_0'. The request is a GET method to the URL 'https://es.wikipedia.org/w/index.php?title=Special:Buscador&search=Cine&prov=acrwl_0'. The status code is 302 Found, and the remote address is 208.80.153.224:443. The response headers show 'Content-Type: text/html; charset=UTF-8' and 'Location: https://es.wikipedia.org/wiki/Cine'. The response body is not visible.

3. Recargar la pagina



The screenshot shows the Wikipedia page for 'Cine' after a refresh. The content is identical to the previous screenshot, but the 'Cine' box now shows a '304 Not Modified' status, indicating that the browser has cached the page and does not need to re-download it. The text and image remain the same.



The screenshot shows the Chrome DevTools Network tab after a page refresh. The selected request is 'Cine', which is a GET method to the URL 'https://es.wikipedia.org/wiki/Cine'. The status code is 304 Not Modified, and the remote address is 208.80.153.224:443. The response headers show 'Accept-Range: bytes', 'Age: 23891', 'Cache-Control: private, s-maxage=0, max-age=0, must-revalidate, no-transform', 'Content-Encoding: gzip', 'Content-Language: es', and 'Content-Security-Policy-Report-Only: script-src 'unsafe-eval' blob: 'self' meta.wikimedia.org *.wikimedia.org *.wikipedia.org *.wikinews.org *.wiktionary.org *.wikibooks.org *.wikiversity.org *.wikisource.org wikisource.org *.wikiquote.org'.

- ¿El servidor recuerda la petición anterior?
- No, porque es un protocolo sin estado (stateless).
- ¿Qué mecanismo permite mantener sesiones de usuario?

Las “Cookies”, ya que el servidor envía un ID de sesión y es recibido por el navegador mediante la cookie.

6 Exploración del protocolo HTTP con cURL

1) Visualizar una solicitud HTTP basica

Ejecutar:

```
curl -v http://neverssl.com
```

Observar:

- Línea de petición (GET)
- Versión HTTP
- Código de estado
- Encabezados de respuesta

```
> curl -v http://neverssl.com
* Host neverssl.com:80 was resolved.
* IPv6: 2600:1f13:37c:1400:ba21:7165:5fc7:736e
* IPv4: 34.223.124.45
* Trying [2600:1f13:37c:1400:ba21:7165:5fc7:736e]:80 ...
* Immediate connect fail for 2600:1f13:37c:1400:ba21:7165:5fc7:736e: La red es inaccesible
* Trying 34.223.124.45:80 ...
* Connected to neverssl.com (34.223.124.45) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: neverssl.com
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Wed, 25 Feb 2026 00:09:32 GMT
< Server: Apache/2.4.62 ()
< Upgrade: h2,h2c
< Connection: Upgrade
< Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT
< ETag: "f79-5e28b29d38e93"
< Accept-Ranges: bytes
< Content-Length: 3961
< Vary: Accept-Encoding
< Content-Type: text/html; charset=UTF-8
```

Responder:

- ¿Qué método HTTP utilizo curl por defecto?

Usa el método GET.

- ¿Qué código de estado devolvió el servidor?

Devolvió un código de estado 200 (OK).

- ¿Qué encabezado indica el tipo de contenido?

El encabezado de respuesta (Response Header).

2) Realizar una solicitud HEAD

Ejecutar:

`curl -I http://neverssl.com`

```
> curl -I http://neverssl.com
<html>
  <head>
    <title>NeverSSL - Connecting ... </title>
    <style>
      body {
        font-family: Montserrat, helvetica, arial, sans-serif;
        font-size: 16px;
        color: #444444;
        margin: 0;
      }
      h2 {
        font-weight: 700;
        font-size: 1.6em;
        margin-top: 30px;
      }
      p {
        line-height: 1.6em;
      }
      .container {
        max-width: 650px;
        margin: 20px auto 20px auto;
        padding-left: 15px;
        padding-right: 15px;
      }
      .header {
        background-color: #42C0FD;
        color: #FFFFFF;
        padding: 10px 0 10px 0;
        font-size: 2.2em;
      }
      .notice {
        background-color: red;
        color: white;
        padding: 10px 0 10px 0;
        font-size: 1.25em;
        animation: flash 4s infinite;
      }
      @keyframes flash {
```

- ¿Qué diferencia hay respecto a la solicitud anterior?

Que no muestra la información sobre la comunicación entre el cliente y el servidor durante la solicitud HTTP.

- ¿Se recibe cuerpo de respuesta?

No.

- ¿Para qué podría servir este método?

Para obtener el recurso directo sin tener información adicional.

Concepto:

- HEAD permite consultar metadatos sin descargar el recurso completo

3) Enviar parámetros en la URL (Query String)

Ejecutar:

```
curl -v https://httpbin.org/get?nombre=Juan&curso=Web
```

Observar:

- Cómo el servidor recibe los parámetros
- Cómo se reflejan en la respuesta

```
> curl -v "https://httpbin.org/get?nombre=Juan&curso=Web"
* Host httpbin.org:443 was resolved.
* IPv6: (none)
* IPv4: 3.93.97.64, 44.195.71.76, 52.6.202.211, 52.200.102.227, 52.202.125.113, 3.94.221.249
* Trying 3.93.97.64:443...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CAPath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.2 (IN), TLS handshake, Certificate (11):
* TLSv1.2 (IN), TLS handshake, Server key exchange (12):
* TLSv1.2 (IN), TLS handshake, Server finished (14):
* TLSv1.2 (OUT), TLS handshake, Client key exchange (16):
* TLSv1.2 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.2 (OUT), TLS handshake, Finished (20):
* TLSv1.2 (IN), TLS handshake, Finished (20):
* SSL connection using TLSv1.2 / ECDHE-RSA-AES128-GCM-SHA256 / secp256r1 / rsaEncryption
* ALPN: server accepted h2
* Server certificate:
* subject: CN=httpbin.org
* start date: Jul 20 00:00:00 2025 GMT
* expire date: Aug 17 23:59:59 2026 GMT
* subjectAltName: host "httpbin.org" matched cert's "httpbin.org"
* issuer: C=US; O=Amazon; CN=Amazon RSA 2048 M03
* SSL certificate verify ok.
* Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 1: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 2: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Connected to httpbin.org (3.93.97.64) port 443
* using HTTP/2
* [HTTP/2] [1] OPENED stream for https://httpbin.org/get?nombre=Juan&curso=Web
* [HTTP/2] [1] [:method: GET]
* [HTTP/2] [1] [:scheme: https]
* [HTTP/2] [1] [:authority: httpbin.org]
* [HTTP/2] [1] [:path: /get?nombre=Juan&curso=Web]
* [HTTP/2] [1] [user-agent: curl/8.14.1]
* [HTTP/2] [1] [accept: */*]
> GET /get?nombre=Juan&curso=Web HTTP/2
```

Responder:

- ¿Dónde viajan los parámetros?

En la URL, específicamente en el “query string” (después del signo ?).

- ¿Qué riesgo implica enviar datos sensibles por URL?

Pueden quedar almacenados en el historial de navegador, puede guardarse en texto plano en el servidor además que son visibles para el usuario.

4) Enviar una solicitud POST

Ejecutar:

```
curl -v -X POST https://httpbin.org/post -d
"usuario=admin&password=1234"
```

Observar:

- Método POST
- Encabezado Content-Type
- Cuerpo de la solicitud

```
> curl -v -X POST https://httpbin.org/post -d "usuario=admin&password=1234"
Note: Unnecessary use of -X or --request, POST is already inferred.
* Host httpbin.org:443 was resolved.
* IPv6: (none)
* IPv4: 52.200.102.227, 3.94.221.249, 52.202.125.113, 44.195.71.76, 52.6.202.211, 3.93.97.64
* Trying 52.200.102.227:443...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CAPath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.2 (IN), TLS handshake, Certificate (11):
* TLSv1.2 (IN), TLS handshake, Server key exchange (12):
* TLSv1.2 (IN), TLS handshake, Server finished (14):
* TLSv1.2 (OUT), TLS handshake, Client key exchange (16):
* TLSv1.2 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.2 (OUT), TLS handshake, Finished (20):
* TLSv1.2 (IN), TLS handshake, Finished (20):
* SSL connection using TLSv1.2 / ECDHE-RSA-AES128-GCM-SHA256 / secp256r1 / rsaEncryption
* ALPN: server accepted h2
* Server certificate:
* subject: CN=httpbin.org
* start date: Jul 20 00:00:00 2025 GMT
* expire date: Aug 17 23:59:59 2026 GMT
* subjectAltName: host "httpbin.org" matched cert's "httpbin.org"
* issuer: C=US; O=Amazon; CN=Amazon RSA 2048 M03
* SSL certificate verify ok.
* Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 1: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 2: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Connected to httpbin.org (52.200.102.227) port 443
* using HTTP/2
* [HTTP/2] [1] OPENED stream for https://httpbin.org/post
* [HTTP/2] [1] [:method: POST]
* [HTTP/2] [1] [:scheme: https]
* [HTTP/2] [1] [:authority: httpbin.org]
* [HTTP/2] [1] [:path: /post]
* [HTTP/2] [1] [user-agent: curl/8.14.1]
* [HTTP/2] [1] [accept: */*]
```

Responder:

- ¿Dónde viajan ahora los datos?

Viajan en el cuerpo (body) de la petición HTTP, no en la URL.

- ¿Se observan en la URL?

No, no se observan en la URL.

- ¿Son visibles en texto plano?

No para alguien interceptando la red, mientras que el servidor que los recibe si los puede ver.

5) Modificar encabezados manualmente

Ejecutar:

```
curl -v https://httpbin.org/headers -H "User-Agent: MiClientePersonalizado"
```

Observar:

- Cómo el servidor recibe el User-Agent

```
> curl -v https://httpbin.org/headers -H "User-Agent: MiClientePersonalizado"
* Host httpbin.org:443 was resolved.
* IPv6: (none)
* IPv4: 52.200.102.227, 3.94.221.249, 52.202.125.113, 44.195.71.76, 52.6.202.211, 3.93.97.64
* Trying 52.200.102.227:443...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CAPath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.2 (IN), TLS handshake, Certificate (11):
* TLSv1.2 (IN), TLS handshake, Server key exchange (12):
* TLSv1.2 (IN), TLS handshake, Server finished (14):
* TLSv1.2 (OUT), TLS handshake, Client key exchange (16):
* TLSv1.2 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.2 (OUT), TLS handshake, Finished (20):
* TLSv1.2 (IN), TLS handshake, Finished (20):
* SSL connection using TLSv1.2 / ECDHE-RSA-AES128-GCM-SHA256 / secp256r1 / rsaEncryption
* ALPN: server accepted h2
* Server certificate:
* subject: CN=httpbin.org
* start date: Jul 20 00:00:00 2025 GMT
* expire date: Aug 17 23:59:59 2026 GMT
* subjectAltName: host "httpbin.org" matched cert's "httpbin.org"
* issuer: C=US; O=Amazon; CN=Amazon RSA 2048 M03
* SSL certificate verify ok.
* Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 1: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 2: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Connected to httpbin.org (52.200.102.227) port 443
* using HTTP/2
* [HTTP/2] [1] OPENED stream for https://httpbin.org/headers
* [HTTP/2] [1] [:method: GET]
* [HTTP/2] [1] [:scheme: https]
* [HTTP/2] [1] [:authority: httpbin.org]
* [HTTP/2] [1] [:path: /headers]
* [HTTP/2] [1] [:accept: */*]
* [HTTP/2] [1] [user-agent: MiClientePersonalizado]
> GET /headers HTTP/2
```

Responder:

- ¿Es posible suplantar información del cliente?

Si, en este caso se ha modificado el User-Agent.

- ¿Qué implicaciones tiene esto en seguridad?

Que el cliente puede alterar/modificar datos que envía hacia el servidor.

6) Simular error 404

Ejecutar:

`curl -v https://httpbin.org/status/404`

```
> curl -v https://httpbin.org/status/404
* Host httpbin.org:443 was resolved.
* IPv6: (none)
* IPv4: 52.200.102.227, 3.94.221.249, 52.202.125.113, 44.195.71.76, 52.6.202.211, 3.93.97.64
* Trying 52.200.102.227:443 ...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.2 (IN), TLS handshake, Certificate (11):
* TLSv1.2 (IN), TLS handshake, Server key exchange (12):
* TLSv1.2 (IN), TLS handshake, Server finished (14):
* TLSv1.2 (OUT), TLS handshake, Client key exchange (16):
* TLSv1.2 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.2 (OUT), TLS handshake, Finished (20):
* TLSv1.2 (IN), TLS handshake, Finished (20):
* SSL connection using TLSv1.2 / ECDHE-RSA-AES128-GCM-SHA256 / secp256r1 / rsaEncryption
* ALPN: server accepted h2
* Server certificate:
* subject: CN=httpbin.org
* start date: Jul 20 00:00:00 2025 GMT
* expire date: Aug 17 23:59:59 2026 GMT
* subjectAltName: host "httpbin.org" matched cert's "httpbin.org"
* issuer: C=US; O=Amazon; CN=Amazon RSA 2048 M03
* SSL certificate verify ok.
* Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 1: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 2: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
* Connected to httpbin.org (52.200.102.227) port 443
* using HTTP/2
* [HTTP/2] [1] OPENED stream for https://httpbin.org/status/404
* [HTTP/2] [1] [:method: GET]
* [HTTP/2] [1] [:scheme: https]
* [HTTP/2] [1] [:authority: httpbin.org]
* [HTTP/2] [1] [:path: /status/404]
* [HTTP/2] [1] [user-agent: curl/8.14.1]
* [HTTP/2] [1] [accept: */*]
> GET /status/404 HTTP/2
```

Responder:

- ¿Qué significa el código 404?

Significa que indica que el servidor no ha encontrado la página o recurso solicitado, es decir, aunque el servidor responde correctamente, el archivo, página o contenido especificado en la URL no existe en el servidor o ha sido eliminado.

- ¿Qué diferencias hay entre 404 y 403?
- Error 404 (Not Found): Indica que la página o recurso solicitado no existe en el servidor.
- Error 403 (Forbidden): Indica que el recurso existe, pero el servidor se niega a permitir el acceso a él.

7) Comparar HTTP y HTTPS

Ejecutar:

```
curl -v http://neverssl.com
```

```
> curl -v http://neverssl.com
* Host neverssl.com:80 was resolved.
* IPv6: 2600:1f13:37c:1400:ba21:7165:5fc7:736e
* IPv4: 34.223.124.45
* Trying [2600:1f13:37c:1400:ba21:7165:5fc7:736e]:80...
* Immediate connect fail for 2600:1f13:37c:1400:ba21:7165:5fc7:736e: La red es inaccesible
* Trying 34.223.124.45:80...
* Connected to neverssl.com (34.223.124.45) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: neverssl.com
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Thu, 26 Feb 2026 00:47:04 GMT
< Server: Apache/2.4.62 ()
< Upgrade: h2,h2c
< Connection: Upgrade
< Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT
< ETag: "f79-5e28b29d38e93"
< Accept-Ranges: bytes
< Content-Length: 3961
< Vary: Accept-Encoding
< Content-Type: text/html; charset=UTF-8
<
<html>
  <head>
    <title>NeverSSL - Connecting ... </title>
    <style>
      body {
        font-family: Montserrat, helvetica, arial, sans-serif;
        font-size: 16px;
        color: #444444;
        margin: 0;
      }
      h2 {
        font-weight: 700;
```

y luego:

```
curl -v https://example.com
```

```
> curl -v https://example.com
* Host example.com:443 was resolved.
* IPv6: 2606:4700::6812:1b78, 2606:4700::6812:1a78
* IPv4: 104.18.27.120, 104.18.26.120
* Trying [2606:4700::6812:1b78]:443...
* Immediate connect fail for 2606:4700::6812:1b78: La red es inaccesible
* Trying [2606:4700::6812:1a78]:443...
* Immediate connect fail for 2606:4700::6812:1a78: La red es inaccesible
* Trying 104.18.27.120:443...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CAPath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519MLKEM768 / id-ecPublicKey
* ALPN: server accepted h2
* Server certificate:
* subject: CN=example.com
* start date: Feb 13 18:53:48 2026 GMT
* expire date: May 14 18:57:50 2026 GMT
* subjectAltName: host "example.com" matched cert's "example.com"
* issuer: C=US; O=SSL Corporation; CN=Cloudflare TLS Issuing ECC CA 3
* SSL certificate verify ok.
* Certificate level 0: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using ecdsa-with-SHA256
* Certificate level 1: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using ecdsa-with-SHA384
* Certificate level 2: Public key type EC/secp384r1 (384/192 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 3: Public key type RSA (2048/112 Bits/secBits), signed using sha1WithRSAEncryption
* Connected to example.com (104.18.27.120) port 443
* using HTTP/2
* [HTTP/2] [1] OPENED stream for https://example.com/
* [HTTP/2] [1] [:method: GET]
* [HTTP/2] [1] [:scheme: https]
```

Identificar:

- Puerto utilizado: Se usaron el puerto 80 (para http) y el 443 (para https).
- Proceso TLS: Solo HTTPS realiza un handshake TLS antes de intercambiar datos.
- Diferencias visibles en la conexión: Uno hace el proceso de cifrado mientras el otro tiene el trafico en texto plano.

7 Preguntas de reflexión final

1. ¿Qué parte de la solicitud controla completamente el cliente?

El cliente controla completamente lo que decide enviar al servidor, es decir, el servidor no controla lo que recibe, solo puede validarlo.

2. ¿Puede el cliente modificar los encabezados?

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Si.

3. ¿Qué sucede si el servidor confía ciegamente en los datos recibidos?

El servidor puede ser víctima de los siguientes ataques:

- Inyección SQL
- XSS
- Escalada de privilegios
- Manipulación de roles
- Bypass de autenticación

4. ¿Qué significa que HTTP sea sin estado al observar múltiples solicitudes con curl?

Significa que no guarda sesiones por defecto, tampoco sabe quién eres al realizar cada solicitud y que no recuerda acciones previas.

8 Paradigma solicitud–respuesta

El paradigma solicitud–respuesta es un modelo de comunicación donde:

1. El cliente envía una solicitud (request) al servidor.
2. El servidor procesa esa solicitud.
3. El servidor devuelve una respuesta (response).

9 ¿Por qué HTTP es considerado sin estado?

- Cada solicitud es independiente.
- El servidor no recuerda solicitudes anteriores.
- No mantiene información automáticamente entre peticiones.

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